DATA MINING & MACHINE LEARNING

Case Studies

Model comparison, evaluation and results on three real-world problems



CASE STUDY 1

Grocery Product Recommendation System



CASE STUDY 2

Sentiment Analysis of Delivery Agent Reviews



CASE STUDY 3

Lung Cancer / Disease Risk Detection

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CASE STUDY · PART 1 OF 3 · OVERVIEW

Grocery Product Recommendation System

PROBLEM STATEMENT

Grocery customers buy products together in patterns that are hard to spot by hand. This project finds frequently bought combinations and recommends related products to improve cross-selling.

OBJECTIVES

1. Find products often bought together
2. Generate association rules (support, confidence)
3. Recommend related products to customers

ALGORITHMS USED

Apriori ★ BEST
Frequent itemsets and association rules. Main algorithm.

KNN
Recommends from similar baskets (cosine distance).

Item-Based CF
Recommends products similar to those in the basket.

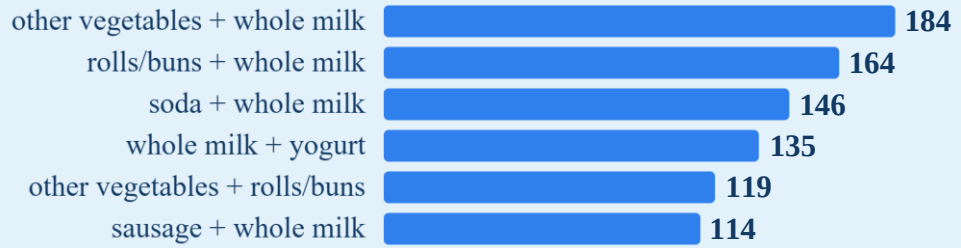
DATASET & SETUP

14758 transactions	**167** unique products
11806 / 2952 train / test (80% / 20%)	**3** algorithms compared

EVALUATION SETUP

- A basket = one customer on one date (2+ products)
- The last product of each test basket is hidden
- Top 5 products are recommended per basket
- All 2,952 test baskets are evaluated

TOP PRODUCT COMBINATIONS (NUMBER OF TRANSACTIONS)



Grocery: Model Evaluation

Three algorithms compared on 2,952 held-out test baskets (top-5 recommendations). Sample values for illustration.

MODEL COMPARISON (TEST SET)

Algorithm	Accuracy	Precision	Recall	F1
★ Apriori	97.12%	84.20%	82.60%	83.39%
KNN	96.96%	72.40%	74.30%	73.34%
Item-Based CF	96.78%	78.10%	79.50%	78.79%

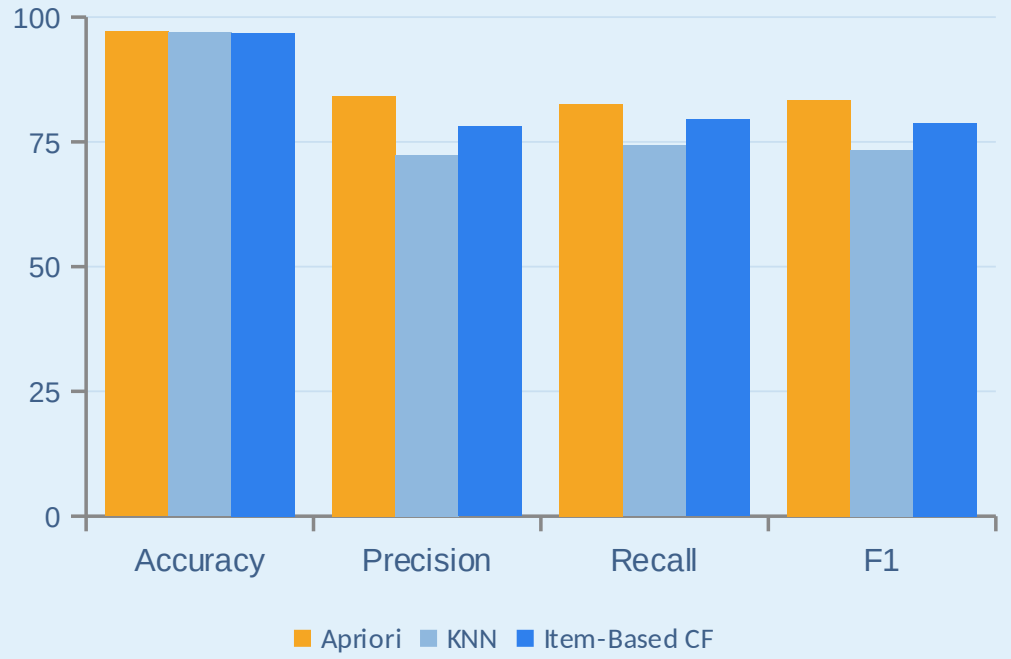
APRIORI · KEY SCORES

0.83 Recall

0.84 Precision

0.83 F1 Score

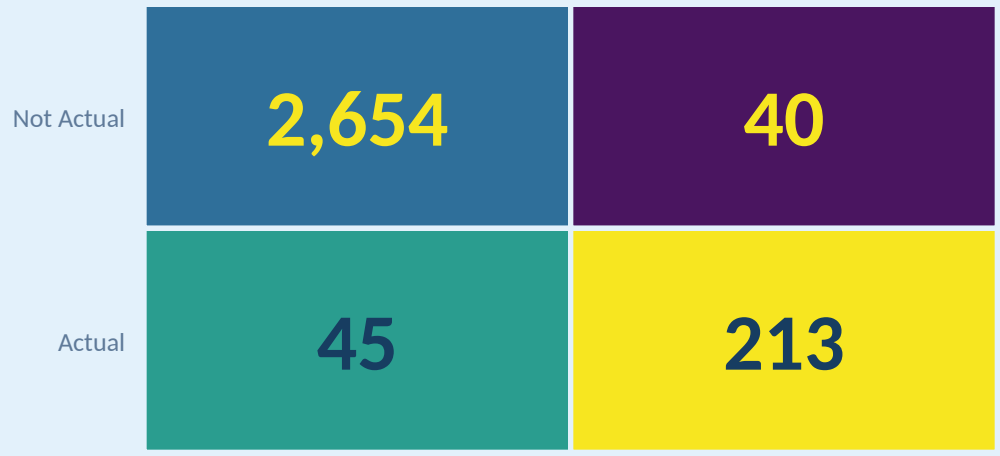
SCORES BY MODEL (%)



★ **BEST: Apriori** – highest accuracy, precision, recall and F1. (Sample values for demonstration.)

Grocery: Results & Final Output

What the best model gets right, where it struggles, and the final result.



2,952 test baskets only (2,654 + 40 + 45 + 213 = 2,952)

INTERPRETATION

- Apriori is best: 97.12% accuracy, 84.20% precision, 82.60% recall, 83.39% F1.
- The matrix uses only the 2,952 test baskets: 2,654 + 40 + 45 + 213 = 2,952.
- Apriori finds 213 of the 258 actual cases (82.6% recall) and makes just 40 wrong recommendations.
- Accuracy is high partly because 2,654 non-matching cases are correctly left out.
- Item-Based CF is second (78.79% F1) and KNN third (73.34% F1).
- Dataset: 14758 transactions (11806 train / 2952 test).

FINAL OUTPUT · BEST MODEL

Apriori · 97.12% accuracy

Recommends the top 5 related products for any grocery item

CASE STUDY 2 · PART 1 OF 3 · OVERVIEW

Sentiment Analysis of Delivery Agent Reviews

PROBLEM STATEMENT

Customer reviews of food delivery hold useful feedback on service quality, but reading them one by one is slow. This project classifies each review as Positive, Neutral or Negative using NLP and machine learning.

OBJECTIVES

1. Classify reviews as Positive, Neutral or Negative
2. Compare three ML models on TF-IDF features
3. Identify the best-performing model

ALGORITHMS USED

Logistic Regression ★ BEST
Linear classifier using class probabilities. Best model.

Linear SVM
Finds the best boundary between the classes.

Multinomial NB
Probabilistic classifier based on word counts.

SENTIMENT LABEL FROM STAR RATING

≤ 2.0 → Negative	≤ 3.5 → Neutral	> 3.5 → Positive

DATASET & SETUP

80,690 Zomato reviews	**3** sentiment classes
64,552 / 16,138 train / test (80% / 20%)	**3** models compared

TEST SET CLASS COUNTS (16,138 REVIEWS)



WORKFLOW

1. **Reading Reviews** — Load Zomato reviews
2. **Text Preprocessing** — Clean text, 3+ words
3. **TF-IDF** — 1–2 word n-grams
4. **ML Model** — Train 3 models
5. **Sentiment Analysis** — Predict sentiment

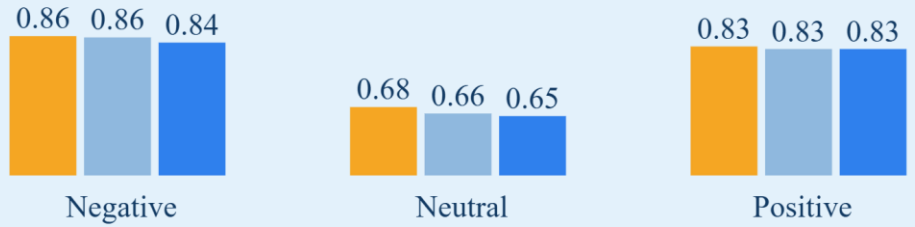
Reviews: Model Evaluation

Three classifiers compared on 16,138 unseen test reviews (macro-averaged scores).

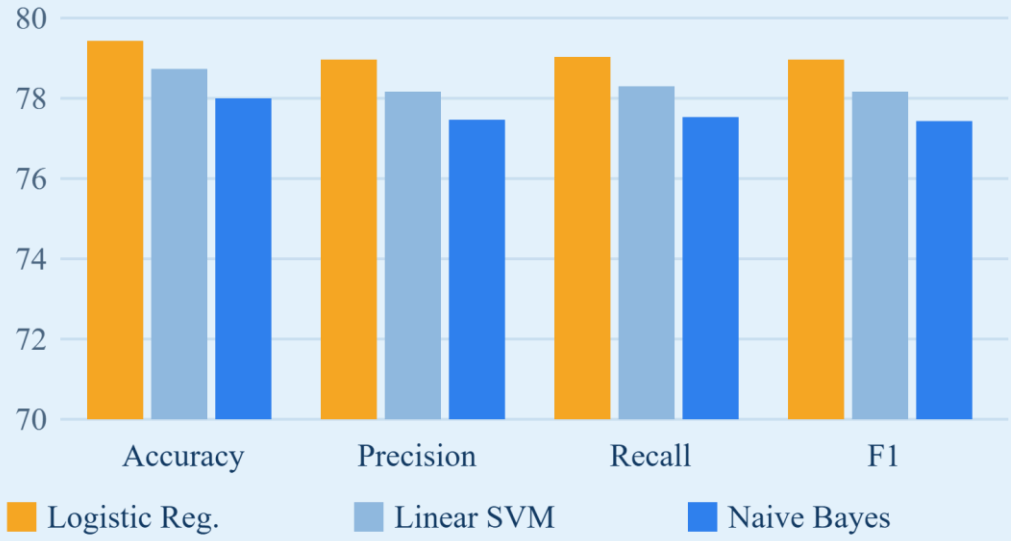
MODEL COMPARISON (TEST SET)

Algorithm	Accuracy	Precision	Recall	F1
★ **Logistic Regression**	79.43%	78.95%	79.05%	78.97%
Linear SVM	78.74%	78.16%	78.30%	78.16%
Multinomial NB	78.01%	77.45%	77.53%	77.43%

F1 PER CLASS · LOGISTIC REG. / LINEAR SVM / NAIVE BAYES



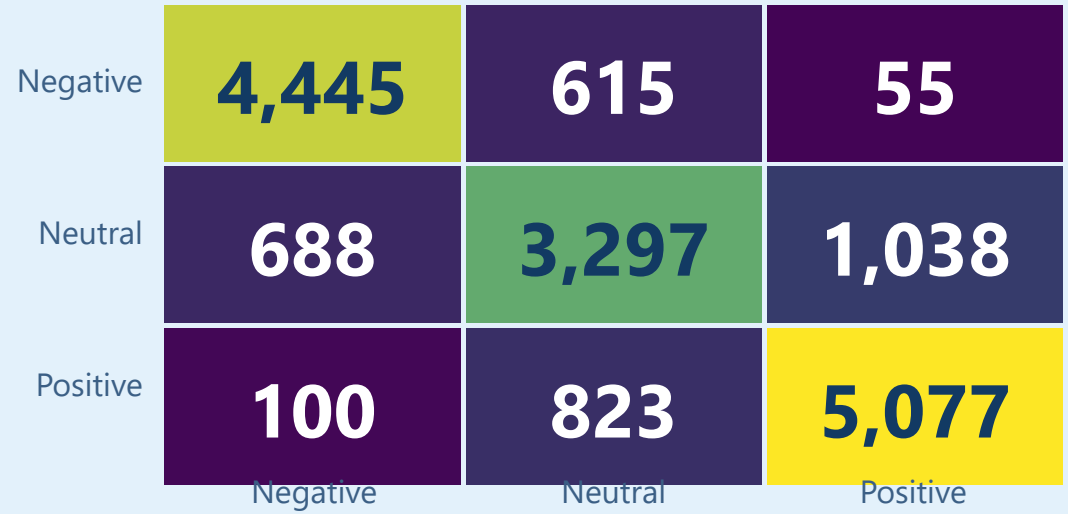
SCORES BY MODEL (%, AXIS STARTS AT 70)



★ **BEST: Logistic Regression** – highest accuracy, precision, recall and F1. Neutral is the hardest class for every model.

Reviews: Results & Final Output

What the best model gets right, where it struggles, and the final result.



Rows: true · Columns: predicted · 16,138 test reviews

INTERPRETATION

- Logistic Regression is best: 79.43% accuracy and 78.97% macro F1.
- Negative and Positive reviews are found best (F1 0.86 and 0.83).
- Neutral is hardest (F1 0.68): 1,038 neutral reviews were predicted Positive and 688 Negative.
- Extreme mistakes are rare: only 55 Negative predicted as Positive and 100 Positive as Negative.
- Dataset: 80,690 Zomato reviews (64,552 train / 16,138 test).

FINAL OUTPUT · BEST MODEL

Logistic Regression · 79.43% accuracy

Predicts each comment as Positive, Neutral or Negative

CASE STUDY 3 · PART 1 OF 3 · OVERVIEW

Lung Cancer Risk Prediction

PROBLEM STATEMENT

Spotting people at higher lung cancer risk early matters, but checking many patient records by hand is slow. This project predicts a Low, Medium or High risk level from health and lifestyle details. It is a dataset exercise, not a diagnosis.

OBJECTIVES

1. Predict Low, Medium or High cancer risk
2. Compare Decision Tree, Logistic Regression and SVM
3. Identify the best-performing model

ALGORITHMS USED

Decision Tree

★ BEST

Splits data with yes/no rules on features. Best F1 in this run.

Logistic Regression

Linear model that predicts risk-level probabilities.

SVM

Finds the best boundary between the risk levels.

DATASET & SETUP

1001
patient records

3
risk levels

800 / 201
train / test (80% / 20%)

3
algorithms compared

RECORDS BY RISK LEVEL (APPROX.)



CASE STUDY 3 · PART 2 OF 3 · EVALUATION

Lung Cancer: Model Evaluation

Three classifiers compared on the 20% test set (values read from the output charts).

MODEL COMPARISON (TEST SET)

Algorithm	Accuracy	Precision	Recall	F1
★ **Decision Tree**	100%	100%	100%	100%
Logistic Regression	91.5%	92.3%	91.5%	91.3%
SVM	88.0%	89.3%	88.0%	87.5%

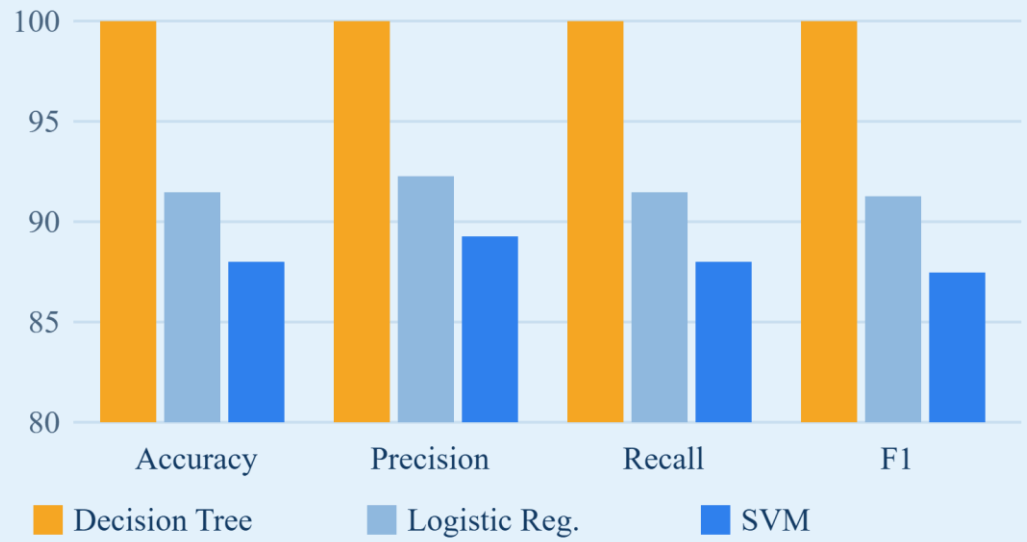
RANKING BY F1 SCORE

100%
1. Decision Tree

91.3%
2. Logistic Reg.

87.5%
3. SVM

SCORES BY MODEL (%, AXIS STARTS AT 80)

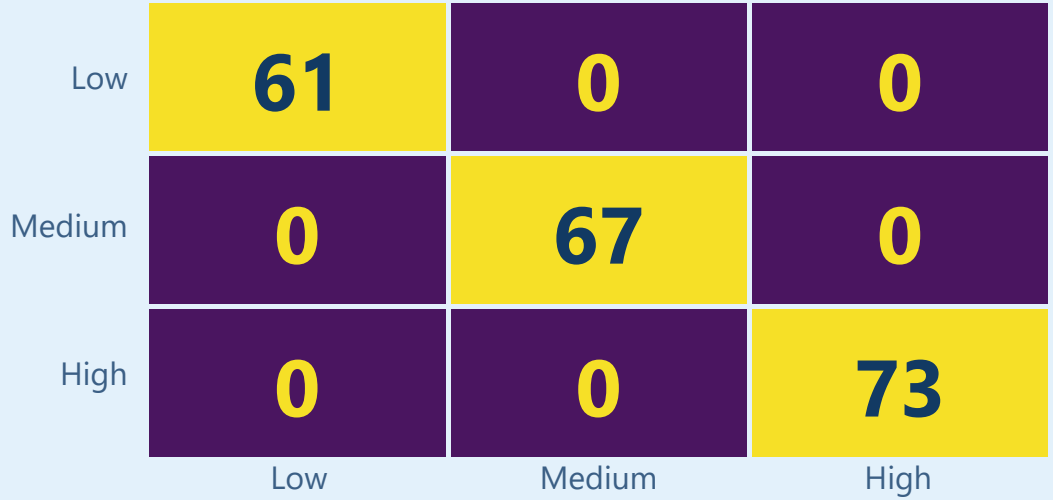


★ **BEST: Decision Tree** – highest on all four metrics. A perfect 100% is unusual, so check the data for features that reveal the risk level.

Lung Cancer: Results & Final Output

How the best model performs on 201 unseen test records, and the final result.

Decision Tree - Confusion Matrix



Rows: true · Columns: predicted · approx. counts, 201 test records

INTERPRETATION

- Decision Tree is best: 100% accuracy, precision, recall and F1, so every test record is on the diagonal.
- Logistic Regression is second (91.5% accuracy) and SVM is third (88%).
- A perfect score is rare. Check for features that give away the risk level before trusting it.
- Risk levels are dataset categories, not confirmed diagnoses.
- Hypothetical scenario only: 5% of 1001 records is 50 people. This is not a cancer forecast.

FINAL OUTPUT · BEST MODEL

Decision Tree · 100% test accuracy

Predicts Low, Medium or High risk from a person's details

Problem 1: Multiclass Classification for Vehicle Type Recognition

PROBLEM STATEMENT

Classify a vehicle as Bike, Bus, Car or Truck from its physical and performance attributes.

DATASET INFORMATION

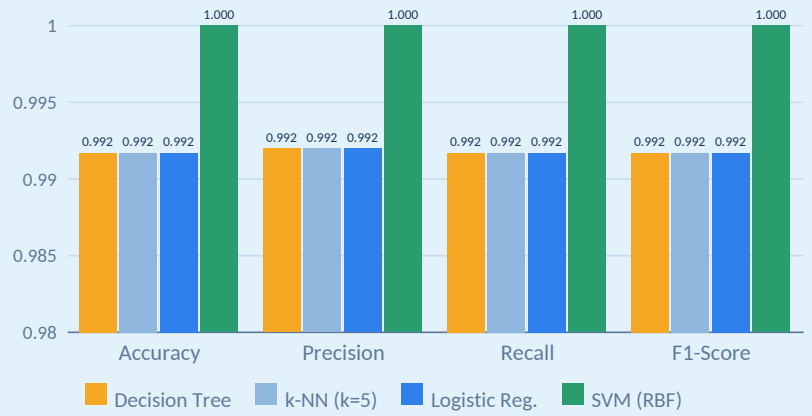
Rows	600
Features	5 (Length, Width, Weight, Engine Capacity, Mileage)
Target	Vehicle_Type
Classes	Bike 144, Bus 113, Car 229, Truck 114
Missing	0
Split	80% train (480) / 20% test (120), stratified

ALGORITHMS USED

- Decision Tree
- k-Nearest Neighbors (k=5)
- Logistic Regression
- SVM (RBF kernel)

Evaluation: hold-out test set (120 samples); scaling for k-NN, LR, SVM

VISUALIZATION – MODEL PERFORMANCE COMPARISON



MODEL PERFORMANCE COMPARISON (TEST SET)

Algorithm	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.9917	0.9920	0.9917	0.9917
k-NN (k=5)	0.9917	0.9920	0.9917	0.9917
Logistic Regression	0.9917	0.9920	0.9917	0.9917
★ **SVM (RBF)**	**1.0000**	**1.0000**	**1.0000**	**1.0000**

KEY METRICS (BEST MODEL)

100% Accuracy	**100%** Precision
100% Recall	**100%** F1-Score

BEST ALGORITHM

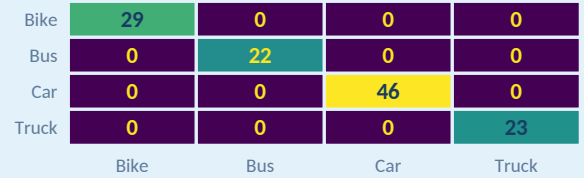
★ **SVM (RBF)**

Accuracy 100% | Weighted F1 = 1.0

INTERPRETATION

- SVM classified all 120 test vehicles correctly; the other three made 1 error (a Bus predicted as Truck).
- Classes differ strongly in size and weight, so near-perfect scores are expected.

CONFUSION MATRIX – SVM (RBF) · 120 TEST VEHICLES



Conclusion: SVM (RBF) is the best model with 100% accuracy and weighted F1 of 1.0; the other three tie at 99.17% (one error each).

Problem 2: Regression Analysis for Student Academic Performance

PROBLEM STATEMENT

Predict a student's final examination score (out of 100) from study habits and academic attributes.

DATASET INFORMATION

Rows	600
Features	5 (Study_Hours, Attendance, Assignment, Internal, Previous)
Target	Final_Score (0–100)
Data types	All numeric (float64)
Missing	0
Split	80% train (480) / 20% test (120)

ALGORITHMS USED

- Linear Regression
- Random Forest Regression (200 trees)

Evaluation: hold-out test set; metrics MAE, MSE, R^2

VISUALIZATION – MODEL PERFORMANCE COMPARISON



MODEL PERFORMANCE COMPARISON (TEST SET)

Model	MAE	MSE	RMSE	R^2
★ **Linear Regression**	**5.774**	**49.692**	**7.049**	**0.518**
Random Forest	6.615	61.013	7.811	0.408

Linear Regression has the lower error on every metric and the higher R^2, so it is selected.

KEY METRICS (BEST MODEL)

0.518 R^2	**5.77** MAE
49.7 MSE	**7.05** RMSE

BEST ALGORITHM

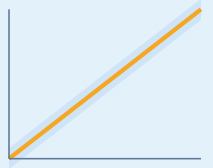
★ **Linear Regression**

R^2 = 0.518 | MAE = 5.77 marks

INTERPRETATION

- Linear Regression explains 52% of variance (R^2 0.518) vs 41% for Random Forest.
- Average error is about 6 marks out of 100; Random Forest likely overfits the 480 training rows.

ACTUAL VS PREDICTED – LINEAR REGRESSION



Perfect-fit line with a ±RMSE (7.05 marks) band: typical predictions land within about 7 marks of the actual score.

Conclusion: Linear Regression predicts better (R^2 0.518, MAE 5.77) than Random Forest (R^2 0.408, MAE 6.62); the data is largely linear.

Problem 3: Disease Risk Classification Using Cross-Validation

PROBLEM STATEMENT

Predict whether a patient is likely to have heart disease from clinical attributes using an SVM with 5-Fold Stratified CV.

DATASET INFORMATION

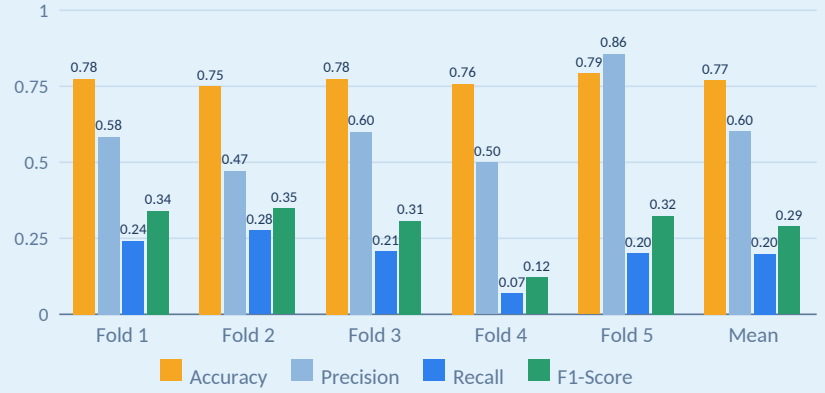
Rows	600
Features	6 (Age, BP, Cholesterol, Max HR, Fasting BS, Chest Pain)
Target	Heart_Disease (1 = Disease)
Classes	No disease 454, Disease 146
Encoding	Chest_Pain_Type one-hot (4 categories)
Missing	0

ALGORITHMS USED

- SVM (RBF kernel)
- StandardScaler + SVM pipeline
- 5-Fold Stratified CV (shuffle, seed 42)

Evaluation: out-of-fold predictions from all 5 folds

VISUALIZATION – MODEL PERFORMANCE COMPARISON



PER-FOLD RESULTS (OUT-OF-FOLD)

Fold	Accuracy	Precision	Recall	F1-Score
Fold 1	0.775	0.583	0.241	0.341
Fold 2	0.750	0.471	0.276	0.348
Fold 3	0.775	0.600	0.207	0.308
Fold 4	0.758	0.500	0.069	0.121
Fold 5	0.792	0.857	0.200	0.324
Mean ± SD	**0.770 ± 0.015**	**0.602 ± 0.136**	**0.199 ± 0.070**	**0.289 ± 0.085**

KEY METRICS (BEST MODEL)

77.0% Accuracy	**60.2%** Precision
19.9% Recall	**28.9%** F1-Score

BEST ALGORITHM

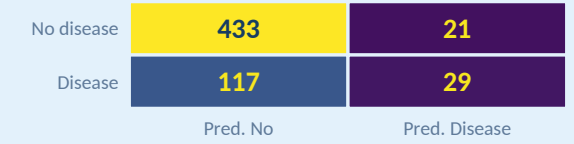
★ **SVM (RBF) + Scaler**

Mean acc. 77.0% | Mean recall 19.9%

INTERPRETATION

- Stable across folds (accuracy SD 1.5 pts) but recall is low in every fold (7–28%).
- Only 29 of 146 disease cases found (117 false negatives); a missed patient is the costly error.

OUT-OF-FOLD CONFUSION MATRIX



600 out-of-fold predictions · rows: true class

Conclusion: The SVM reaches 77% accuracy but only 20% recall: most disease cases are missed, so it is unsuitable for screening as is. Educational data only.

Problem 4: Comparing Regression Models for Employee Productivity

PROBLEM STATEMENT

Predict employee productivity (0–100) from work-related attributes and compare three regression models using cross-validation.

DATASET INFORMATION

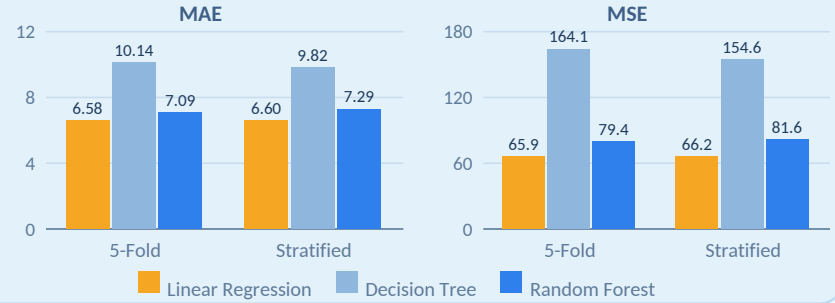
Rows	600
Features	5 (Experience, Training, Tasks, Attendance, Work Hours)
Target	Productivity_Score (0–100)
Top correlate	Tasks_Completed (r = 0.68)
Missing	0
CV	5-Fold and Stratified 5-Fold (target in 5 quantile bins)

ALGORITHMS USED

- Linear Regression
- Decision Tree Regression
- Random Forest Regression (200 trees)

Evaluation: mean ± SD of MAE and MSE over 5 folds

VISUALIZATION – MODEL PERFORMANCE COMPARISON



MODEL COMPARISON (MEAN ± SD)

Model	MAE (5-Fold)	MSE (5-Fold)	MAE (Strat.)	MSE (Strat.)
★ Linear Regression	**6.58 ± 0.34**	**65.9 ± 4.3**	**6.60 ± 0.28**	**66.2 ± 3.1**
Decision Tree	10.14 ± 0.26	164.1 ± 11.7	9.82 ± 0.65	154.6 ± 21.9
Random Forest	7.09 ± 0.47	79.4 ± 9.1	7.29 ± 0.24	81.6 ± 5.0

FOLD-WISE MAE & MSE (5-FOLD CV)

Fold	LR MAE	LR MSE	DT MAE	DT MSE	RF MAE	RF MSE
F1	6.54	65.8	10.00	159.2	7.10	82.5
F2	6.98	71.3	10.04	159.6	7.66	85.7
F3	6.31	60.3	10.34	181.1	6.78	74.4
F4	6.12	62.0	9.79	147.7	6.38	64.5
F5	6.96	70.2	10.52	172.9	7.53	90.1
Mean	**6.58**	**65.9**	**10.14**	**164.1**	**7.09**	**79.4**

KEY METRICS (BEST MODEL)

6.58 MAE	**65.9** MSE
6.60 MAE (Strat.)	**66.2** MSE (Strat.)

BEST ALGORITHM

★ **Linear Regression**

Lowest mean MAE and MSE in both CV schemes

INTERPRETATION

- Mean MAE (5-Fold): Linear 6.58 < Random Forest 7.09 < Decision Tree 10.14.
- Decision Tree overfits (MSE >2x Linear); stratified and plain CV give the same ranking.

Conclusion: Linear Regression has the lowest average error (MAE 6.58, MSE 65.9) under both CV schemes; Random Forest is close, Decision Tree is worst.